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Fly me to the canopy: Diptera communities in oak forest crowns as bioindicators of stand decline

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Abstract

1. Diptera are among the most diverse and ecologically important insect orders in forest ecosystems and hold strong potential as bioindicators of ecosystem transformations under anthropogenic and environmental pressures.
2. We examined the effects of oak decline, a process that alters canopy structure and resource availability, on canopy-dwelling Diptera across 30 oak stands in six French forests. Diptera were sampled using green multi-funnel traps placed in the lower canopy. Decline severity was derived from Sentinel-2 imagery at multiple spatial scales.
3. In total, more than 27,000 specimens were collected, representing 60 families and approximately 500 taxa.
4. Overall diversity increased with decline severity, particularly for saproxylic and floricolous guilds. This pattern may result from changes in microclimatic conditions in more open, declining stands that promote Diptera activity, as well as from increased availability of deadwood and floral resources.
5. Families such as Dolichopodidae, Empididae, Hybotidae and Anthomyiidae responded positively to decline severity, whereas Mycetophilidae and other mushroom-associated taxa declined, possibly due to reduced fungal productivity and drier microhabitats. Stand characteristics, including height, density and oak

† Deceased.

For affiliations refer to page 12

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basal area, generally had negative effects on richness and abundance, potentially through their influence on microclimate or resource distribution in denser stands.
6. Co-occurrence analyses indicated that species associations were predominantly random, followed by positive associations, and this pattern did not vary with decline severity.
7. These results suggest that Diptera may serve as sensitive indicators of stand condition and structural changes associated with climate-driven canopy decline.

KEYWORDS
ecological guilds, floricolous insects, Forest dieback, saproxylic insects, tree-related microhabitats

INTRODUCTION

Diptera constitute a major component of forest biodiversity, yet they remain largely understudied due to their small size, cryptic life histories and the taxonomic challenges associated with species-level identification across many families of this group (Courtney et al., 2017; Ulyshen, 2018). Knowledge on this group can be regarded as fragmentary, and even in Europe thousands of species remain undescribed (Morinière et al., 2019; Chimeno et al., 2022). Despite these limitations, their ecological significance is increasingly acknowledged. Diptera contribute to a wide range of ecosystem functions, including pollination, but their most prominent roles in forest ecosystems relate to nutrient cycling and energy flow: they function as decomposers of various organic substrates, including wood and leaf litter, as predators and parasites of other arthropods and vertebrates, and as key prey for numerous forest-dwelling organisms (Courtney et al., 2017; Ulyshen, 2018). This order displays remarkable diversity among saproxylic species, particularly during the larval stage. Numerous dipteran families and species are tightly associated with microhabitats linked to dead or decaying wood, including sap runs, tree cavities, fungal fruiting bodies and rotting wood (Hövemeyer & Schauermann, 2003; Mlynarek et al., 2018; Ulyshen, 2018). This ecological breadth highlights both the versatility and the high degree of habitat specialization that characterize the group.

Given their diversity and the habitat specificity of certain taxa, several dipteran groups have been identified as valuable bioindicators in various ecosystems, especially streams and wetlands (Chowdhury et al., 2023; Courtney et al., 2017), but also specific habitats (e.g., Montoya et al., 2021). Forest-associated species hold similar potential, particularly because of their sensitivity to forest management practices (Byriel et al., 2020; Courtney et al., 2017; Ulyshen, 2018). Tree and herb canopy cover, as well as herb diversity, are considered major drivers of Diptera communities in temperate forests (Scherber et al., 2014). In addition, forest-dwelling and/or saproxylic flies are known to respond to both the quantity and spatial continuity of deadwood resources and microhabitats at local and landscape scales (Herrault et al., 2016; Oosterbroek et al., 2006; Vanderwel et al., 2006). Some families also appear to be sensitive to the temporal continuity of forests (Herrault et al., 2016; Økland, 1994) and the forest area and connectivity at broader spatial scales (Herrault et al., 2016; Økland, 1996; Ouin et al., 2006). In line with this, several studies also indicated that Diptera communities are sensitive to natural forest disturbances like wildfires (e.g., Kwon et al., 2013; Thompson et al., 2022) and windstorms (e.g., Durska, 2013; Wermelinger et al., 2017). However, studies investigating the response of Diptera communities to drought-driven declines are lacking (Cours et al., 2023).

Droughts and hot spells have been increasing in frequency, intensity and spatial extent across central and southern Europe, with a marked recent acceleration (Carnicer et al., 2011; Senf et al., 2020). These trends are projected to intensify in temperate regions (Samaniego et al., 2018; Spinoni et al., 2018), driving major shifts in forest organization and functioning (Senf et al., 2020). Such disturbances strongly influence the availability of trophic resources and microhabitats for forest arthropods (Cours et al., 2023, 2025), primarily through canopy degradation (Cours et al., 2023; Sallé et al., 2021). Canopy dieback reduces the quantity and quality of foliage and other tree resources (Günthardt-Goerg et al., 2013; Hu et al., 2013; Ishii et al., 2004), but can also increase deadwood and tree-related microhabitats such as fungal fruiting bodies, trunk cavities and sap runs (Bouget et al., 2024; Sallé et al., 2021; Zemlerová et al., 2023). Changes in canopy cover further reshape forest microclimates by altering light regimes and buffering capacities for heat and moisture (Kopáček et al., 2020; Verdonck et al., 2025), while modulating the provision of ground-level resources, such as floral resources (Heil & Burkle, 2018). The effects of drought-driven decline on arthropods have been studied across several taxa and guilds (e.g., Cours et al., 2025; Le Souchu, Cours, et al., 2024; Martel & Mauffette, 1997; Sallé et al., 2020; Stone et al., 2010), showing guild-specific responses (Le Souchu, Cours, et al., 2024; Sallé et al., 2020) but also idiosyncratic variation within taxa (Le Souchu, Cours, et al., 2024; Vincent et al., 2020). Consequently, predicting arthropod responses to drought-driven declines in temperate forest canopies remains challenging (Cours et al., 2023; Sallé et al., 2021). Given their exceptional taxonomic and ecological diversity, Diptera may reveal patterns in community responses that might not be detectable in other taxonomic groups.

We hypothesized that oak decline, through the enhancement of deadwood-related resources and microhabitats associated with decaying trees, would promote cavicolous Diptera (i.e., associated with trunk cavities), species associated with sap exudates, and more generally saproxylic taxa, as previously observed in other taxonomic groups (Cours et al., 2023, 2025). In addition, canopy opening in

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declining stands was expected to favour floricolous species, as has been regularly reported for pollinators following forest disturbances (Le Souchu, Cours, et al., 2024; Viljur et al., 2022). Changes in microclimatic conditions were anticipated to exert contrasting effects. On the one hand, warmer and/or drier summer conditions could benefit some ground-dwelling species, as hypothesized for rhizophagous beetles or ground-nesting Hymenoptera (Le Souchu, Cours, et al., 2024; Sallé et al., 2020). On the other hand, drier summer conditions could negatively impact species associated with moist microhabitats. Likewise, lower tree productivity in declining stands may decrease mushroom biomass (Egli et al., 2010; Olano et al., 2020), thereby cascading onto mycetobiont species. Finally, canopy opening could adversely affect forest-dwelling species. Because such structural and microclimatic modifications are likely to affect insect communities in the canopy layer, we specifically focused on Diptera to assess how oak decline alters this component of forest biodiversity. Based on these expectations, our objectives were threefold: (i) to characterize the Diptera community inhabiting oak canopies, (ii) to assess the relative influence of stand-related variables and forest decline severity across multiple landscape scales on Diptera community structure and (iii) to evaluate the specific responses of Diptera families and ecological guilds to forest decline and stand characteristics.

MATERIALS AND METHODS

Study sites

In 2021, we sampled six oak-dominated forests in France (*Quercus petraea* (Matt.) Liebl. and *Quercus robur* L.; Figure 1, Table S1), all

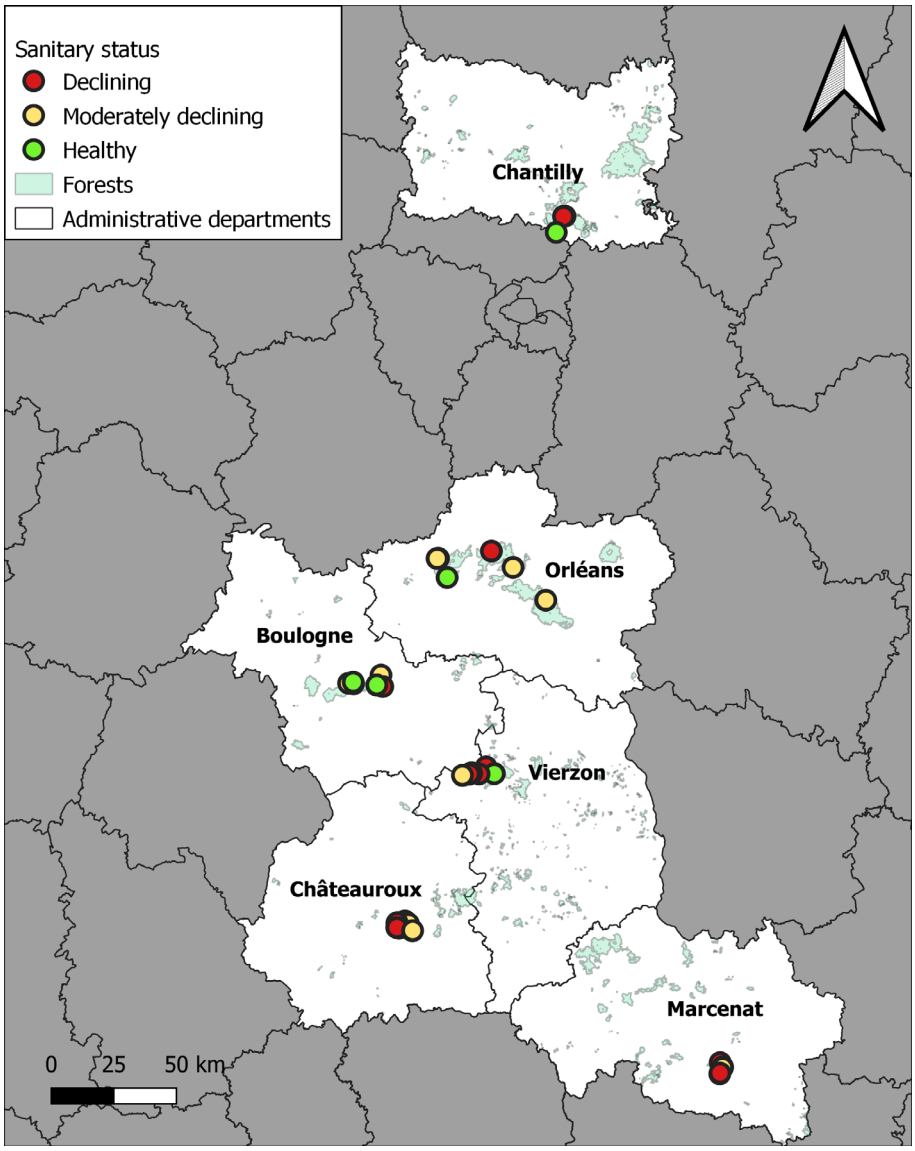


FIGURE 1 Map of the oak forests and stands sampled in 2021. Colours indicate the sanitary status of the stands during the sampling (see Material and methods).

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managed by the French National Forest Service. Four of these sites (Vierzon, Marcenat, Châteauroux and Chantilly) have experienced drought-driven decline, either historically (Vierzon) or more recently following consecutive summer droughts in 2018 and 2019, sometimes exacerbated by unfavourable soil conditions and outbreaks of cockchafers (*Melolontha* spp.; Coleoptera, Scarabaeidae). In contrast, no large-scale forest decline has been reported in the Orléans and Boulogne forests, although some stands showing moderate decline were present locally.

Within each forest, we selected oak-dominated stands exhibiting contrasting levels of decline, identified through local forester expertise and visual assessment (six stands in Orléans, Châteauroux, Vierzon and Boulogne; three stands in Chantilly and Marcenat; Figure 1), aiming to include an equal number of healthy, moderately declining and severely declining stands. In total, 30 stands were sampled, each covering at least 3 ha and separated by a minimum distance of 500 m. In each stand, we first selected two living oaks for trap installation, one healthy and one declining tree, separated by at least 50 m, whenever both conditions were present. The selected trees were located at the centre of two plots in which decline severity and stand characteristics were subsequently quantified. In each stand, we measured stand height with a clinometer (measured on five dominant trees), tree density (one measurement per plot), oak diameter at breast height (DBH; measured on 10 trees) and oak basal area with a relascope (one measurement per plot). Soil pH was assessed from three soil samples per plot. In a subset of 18 stands, deadwood was characterized and quantified, and tree-related microhabitats were visually inventoried. These data are available in Bouget et al. (2024).

Decline severity was quantified using time series of Sentinel-2 imagery of 2020 and 2021 and the RECONFORT model developed by Mouret et al. (2024), applied to three spatial buffers (25, 200 and 500 m) centred around each plot. These spatial scales covered a spatial range previously identified as the most relevant for detecting decline effects on the taxonomic and guild diversity of saproxylic beetles (Cours et al., 2022; Seibold et al., 2017). In brief, map results of oak health status were used at a scale of 10 m and classified into three categories: healthy (less than 20% of declining trees in the plot) coded 1, moderately declining (between 20% and 50% of declining trees in the plot) coded 2 and severely declining (more than 50%) coded 3. Within the buffer zones created with the QGIS 3.28 software, zone statistics were carried out to extract the average value of the pixels present. These average dieback severity values (DSV) obtained within each buffer zone were then categorized as healthy ($DSV < 1.4$), moderately declining ($1.4 < DSV < 2$) and declining ($2 < DSV < 3$) to have an even distribution of plots along a decline severity gradient.

Insect sampling

Insect sampling was conducted using green multi-funnel traps (12-fluon coated funnels; ChemTica Internacional, San José, Costa Rica), suspended in the lower canopy (ca. 10–15 m above ground). Collection cups were filled with a 50% (v/v) monopropylene glycol–water solution supplemented with a drop of detergent. Traps were deployed from late March to mid-August 2021, coinciding with peak insect activity, and emptied monthly. This sampling design was originally conceived to monitor borer communities along an oak decline gradient (Le Souchu, Bouget, & Sallé, 2024). It also proved effective in capturing a broad spectrum of other forest insect taxa (Le Souchu, Cours, et al., 2024; Sallé et al., 2020; Vincent et al., 2020). Although these traps were likely less efficient than Malaise, light, pan, or baited traps commonly used for Diptera (Brown, 2020; Scherber et al., 2014), they nonetheless collected a substantial number of dipteran individuals.

Identification and ecological trait list

All Diptera specimens were initially sorted and identified to the family or superfamily level using the identification key of Oosterbroek (2015). Subsequently, specimens were sent to 26 taxonomic specialists for identification to genus, species, or morphospecies level when possible. Due to the lack of available experts, several families (Cecidomyiidae, Ceratopogonidae, Chironomidae, Ephydridae, Heleomyzidae, Hippoboscidae, Lauxaniidae, Microphoridae, Odiniidae, Phoridae and Sciaridae) were not identified beyond the family level, except for a few isolated specimens.

In some families or genera, available identification keys do not allow the determination of females beyond the genus level, while in other cases, specimens were too damaged during collection or storage to permit identification below family or genus. Most specimens were identified using morphological keys; however, Mycetophilidae, Culicidae and Psychodidae were identified through molecular techniques (see Protocols SD1, SD2 and SD3).

We classified taxa into guild categories based on literature and expert knowledge: (i) silvicolous species, preferentially occurring in forests; (ii) adult guilds: floricolous species, whose adults frequently visit flowers; and larval guilds, including (iii) soil-dwellers, with larvae developing in soil, litter, dung, or other decomposing ground material; (iv) mycetobionts, with larvae developing in the fruiting bodies of soft fungi but not associated with dead wood; (v) cavicolous species, with larvae developing primarily in trunk cavities; (vi) sap-run specialists, with larvae developing preferentially in slime fluxes; and (vii) saproxylic species, with larvae developing in dead or decaying wood, including cavicolous and sap-run specialists. We also noted whether the larvae of detected indicator species (see below) were predatory or parasitoid, although this was not considered a separate guild in our analysis, as the trophic regime of dipteran larvae is frequently unknown.

Statistical analysis

Analyses were conducted in R (version 4.5.1). We first extrapolated the species richness for the whole community and for three decline

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categories. We used only taxonomic families for which more than 90% of the specimens were identified. Species richness was estimated using sample-size-based rarefaction and extrapolation curves implemented in the iNEXT package (Hsieh et al., 2016). Analyses were conducted using abundance data and Hill numbers of order $q = 0$ (species richness). The asymptotic richness estimator corresponded to the Chao1 estimator. Extrapolation was performed up to 30,000 individuals to allow comparison among assemblages. Confidence intervals were obtained by bootstrap resampling implemented in the package.

Second, we assessed how environmental variables affected dipteran community composition. Species abundance data were first filtered to remove very rare taxa (species represented by fewer than 10 individuals) and species showing near-zero variance across samples. Species variance was calculated across samples, and taxa with a variance lower than 0.01 were excluded, as such taxa contribute little to differences in community composition among samples. This filtering step reduces the influence of sparsely represented taxa and improves ordination stability. The resulting species matrix was Hellinger-transformed (square-root transformation of relative species abundances, vegan package, Oksanen et al., 2025) to reduce the influence of dominant taxa, as recommended for community composition analyses (Legendre & Legendre, 2012).

A subset of environmental variables was pre-selected based on ecological relevance and prior correlation analyses to reduce redundancy. This pre-selection limited the set of candidate predictors and avoided including strongly collinear variables. All possible combinations of these variables were then tested in PERMANOVA models (adonis2, Bray–Curtis dissimilarities, 9999 permutations, vegan package), with forest identity included as a stratification factor to account for the nested sampling design. Because likelihood-based information criteria are not directly available for PERMANOVA models, model selection was based on an AIC_c-like approximation derived from the proportion of variance explained (R^2), following the logic of the AICcPermanova framework (Guillot & Rousset, 2013). This approach allowed us to restrict the candidate predictor set based on ecological considerations while still exploring alternative model structures. Variance inflation factors (VIF, car package; Fox & Weisberg, 2019) were subsequently calculated for the best-supported model to confirm that multicollinearity was low (VIF < 5; James et al., 2013). This two-step approach ensures both efficient model selection and robust control of collinearity.

Finally, the relative contribution of each environmental predictor to community dissimilarity was calculated from its proportion of explained variance. Non-metric multidimensional scaling (NMDS, vegan package, $k = 4$, Bray–Curtis index, 1000 permutations) was performed to visualize differences among Diptera communities in healthy, moderately declining and declining plots, based on average DSV (see Study sites).

We used the indicator species analysis (IndVal.g) to identify species associated with a decline category (healthy, intermediate, declining) (indicspecies package; Dufrêne & Legendre, 1997). This method calculates the association value (IndVal index) between a species and a decline category, based on its specificity (the proportion of individuals of the species found in that category) and its fidelity (the proportion of sites within the category where the species occurs). To simplify interpretation, the analysis was restricted to species associated with a single category using the argument duleg = TRUE. For this analysis, species occurring in fewer than 6 traps (i.e., <10% of traps) or represented by fewer than 20 individuals were removed. This step was applied to reduce noise from extremely rare or infrequently encountered taxa, which can obscure the detection of significant indicator associations. The statistical significance of each association was tested with a permutation test (9999 permutations).

To assess the co-occurrence pattern in the community of forest Diptera, we used the R package cooccur (function *cooccur*, presence-absence matrix, Griffith et al., 2016) for the entire Diptera community and for three decline categories (estimated as above within a 500 m radius according to the average decline severity values (DSV); healthy: DSV < 1.4, $n = 19$, moderately declining: 1.4 < DSV < 2, $n = 19$, declining: 2 < DSV < 3, $n = 22$). Species pairs that are expected to share less than 1 site were removed from the analysis (Griffith et al., 2016). The observed frequency of co-occurrence can be significantly greater than expected (positive association), significantly smaller than expected (negative association), or not significantly different than the expectation (random association) (Griffith et al., 2016; Veech, 2013).

To evaluate the influence of environmental variables on the abundance and richness of each Diptera family and ecological guild, we used a model selection framework based on generalized linear mixed models (GLMMs, glmmTMB package; McGillycuddy et al., 2025). The analyses were restricted to families and guilds represented by more than 50 individuals to ensure that sample sizes were sufficient for reliable statistical inference. Additionally, families for which fewer than 90% of specimens could be identified were excluded from species richness analyses to avoid bias due to incomplete taxonomic resolution. These thresholds were chosen to balance the inclusion of taxa while maintaining the robustness and interpretability of the results. For each response variable, we first identified the most appropriate error distribution (Poisson, negative binomial, log-normal or normal) using likelihood-based fit statistics (fitdistrplus package; Delignette-Muller & Dutang, 2015). As potential predictors, we considered the three mean decline severity values measured at different spatial scales (25, 200, and 500 m), as well as stand and environmental descriptors (pH, stand height, oak DBH, tree density and oak basal area). To reduce redundancy among the three highly correlated decline variables, we first identified the most explanatory one for each response variable, based on AIC_c values. Candidate models were generated by combining one to three fixed effects among the measured predictors (soil pH, stand height, tree density, oak DBH, oak basal area and decline severity across three spatial scales), while including a hierarchical random-effect structure (stand nested within forest). To limit multicollinearity, fixed-effect sets with VIFs greater than 5 were excluded (James et al., 2013). For each valid candidate model, we calculated the AIC_c (MuMIn package; Bartoń, 2025). The model with the lowest AIC_c was retained as the best candidate and compared with a null model (intercept + random effects only). Following Burnham and Anderson (2002), only models with AIC_c values at least two points lower than the null ($\Delta AIC_c \geq 2$) were considered informative. Non-informative models were excluded from the graphical outputs.

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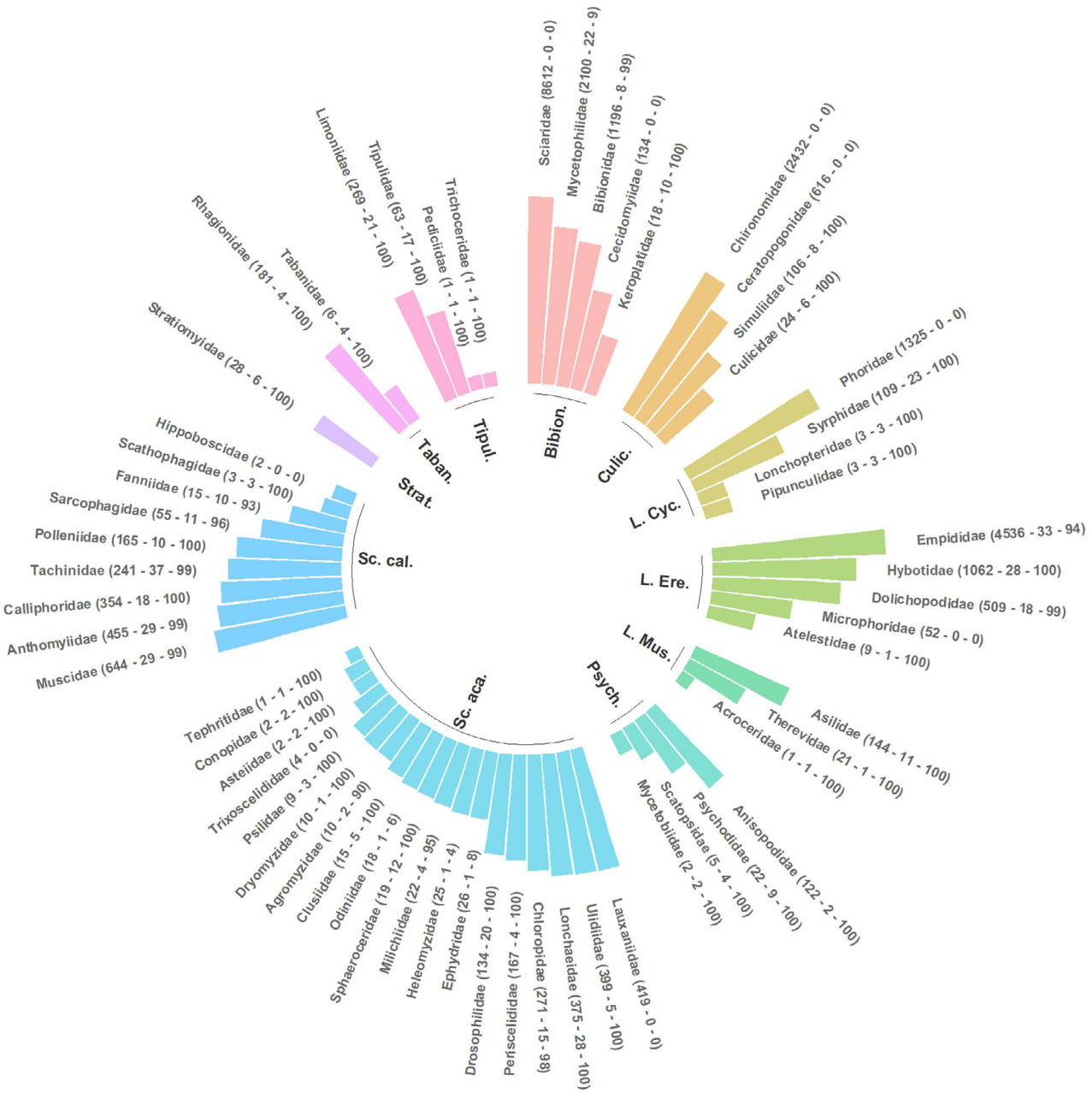


FIGURE 2 Log-transformed abundance of the Diptera families collected in the oak canopies. Families are arranged into 11 infra-orders or clades: Bibionomorpha (Bibion.), Culicomorpha (Culic.), Lower Cyclorrhapha (L. Cyc.), Lower Eremoneura (L. Ere.), Lower Muscomorpha (L. Mus.), Psychodomorpha (Psych.), Schizophora acalyptrate (Sc. aca.), Schizophora calyptrate (Sc. cal.), Stratiomyomorpha (Strat.), Tabanomorpha (Taban.), Tipulomorpha (Tipul.). Numbers between parentheses indicate the number of specimens, number of taxa identified (at the species or genus level) and the percentage of individuals identified.

RESULTS

Community composition and diversity

We collected and identified a total of 27,782 specimens belonging to 60 dipteran families (Figure 2). The most abundant families were Sciaridae (8612 individuals), Empididae (4536), Chironomidae (2432), Mycetophilidae (2100), Phoridae (1325), Bibionidae (1196) and Hybotidae (1062). Overall, 42% of the specimens were identified to genus or species level, resulting in the recognition of 500 taxa. There were 203 singletons and 77 doubletons. Overall, 138 taxa were represented by at least 10 individuals. For Mycetophilidae, amplification frequently failed, probably due to DNA degradation in the samples, and only 9% of specimens were successfully barcoded; of these, 21 were identified to species level and 4 to genus level. Among families with successfully identified taxa, the most diverse were Tachinidae (37 taxa), Empididae (33), Anthomyiidae (29) and Muscidae (29).

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The rarefaction curve for the whole community barely approached an asymptote (Figure 3), suggesting that approximately 600 fly species were present in the community. Declining plots were predicted to harbour more taxa than healthy and moderately declining ones (Figure 3).

Effects of decline severity and other environmental variables on the community

To evaluate the relative contribution of environmental variables to inter-plot variance in the dipteran community, we first identified the most relevant predictors. The decline-related variables (DSV within 25, 200 and 500 m) were strongly correlated ($R^2 = 0.59–0.80$); therefore, only the mean DSV within 500 m was retained, as it explained the largest share of variance in community composition. Model selection based on AIC_c identified the best model as including DSV within 500 m, stand height and oak basal area (Table 1). Among these, decline severity emerged as the strongest driver and was the only variable that significantly contributed to community structuring (Table 1). According to the NMDS ordination, communities associated with healthy and severely declining stands showed limited overlap, while communities in moderately declining stands overlapped with both extremes. (Figure 4).

The indicator species analysis showed that 15 species were significantly associated with declining stands, most of them belonging to the Dolichopodidae, Empididae and Hybotidae families (Table 2). These species were frequently floricolous, predatory and/or saproxylic. In contrast, only one species of Tachinidae, *Prosethilla kramerella* (Stein), was associated with healthy stands (Table 2).

Effects of decline severity and other environmental variables on the taxonomic families

Eight families (Anthomyiidae, Asilidae, Ceratopogonidae, Chloropidae, Dolichopodidae, Empididae, Hybotidae and Lauxaniidae) were more abundant under higher decline severity, and among the families identified to species or genus level, four were also more diverse (Dolichopodidae, Empididae, Periscelididae and Psychodidae) (Figure 5). Two additional families showed a tendency towards higher

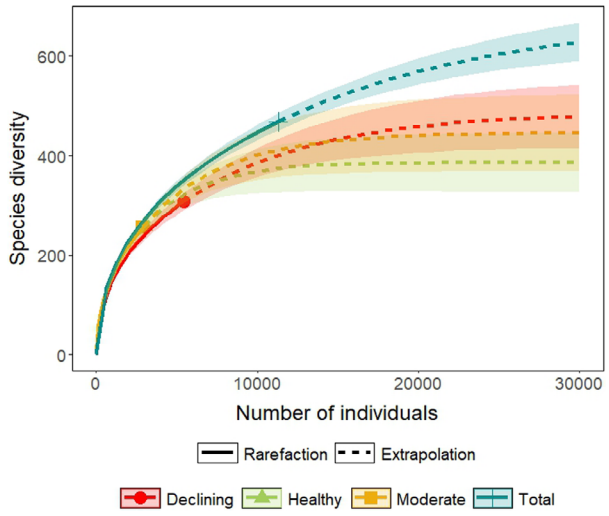


FIGURE 3 Rarefaction curves for the entire dataset, and by decline category (estimated within a 500 m radius according to the average decline severity values [DSV]; healthy: DSV < 1.4, $n = 19$, moderately declining: 1.4 < DSV < 2, $n = 19$, declining: 2 < DSV < 3, $n = 22$). Families that were not or partially identified at lower taxonomic levels were excluded from the analyses.

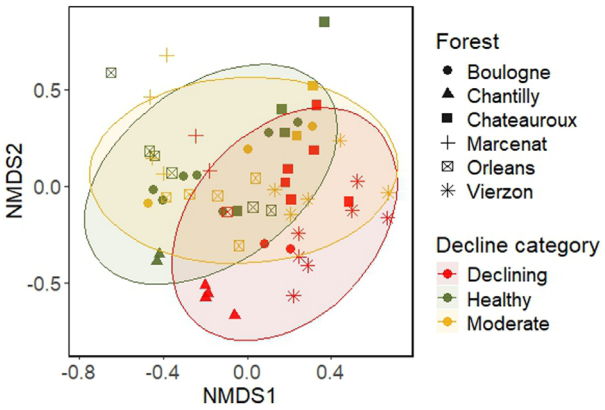


FIGURE 4 NMDS ($k = 4$, stress = 0.17, ellipses: 0.9) for Diptera communities in each trap (i.e., per plot), grouped by decline category (estimated within a 500 m radius according to the average decline severity values (DSV); healthy: DSV < 1.4, $n = 19$, moderately declining: 1.4 < DSV < 2, $n = 19$, declining: 2 < DSV < 3, $n = 22$). Only the two first dimensions are shown on the figure.

TABLE 1 Results of the permutational analysis of variance assessing the effects of mean decline severity values (DSV) over 500 m around each trap, stand height and oak basal area on the community of Diptera trapped in the canopy of several French oak forests.

Fixed effect	df	Sum of squares	R^2	Pseudo-*F*	*p*-value
Mean decline severity (500 m)	1	1.157	0.081	5.423	0.0001
Stand height	1	0.534	0.037	2.500	0.2307
Oak basal area	1	0.543	0.037	2.545	0.1858
Residuals	56	11.950	0.833		

Note: Permutations were stratified within forests (see Material and methods).

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TABLE 2 Indicator values (IndVal) of Diptera species significantly associated with a category of oak decline (healthy, intermediate and declining, estimated within a 500 m radius according to the average decline severity values—see Material and methods for details).

Family	Species	Flor.	Pred.	Sx.	IndVal	*p*-value	Decline category
Asilidae	*Tolmerus atricapillus*				0.56	**	Declining
Dolichopodidae	*Medetera dendrobaena*				0.47	*	Declining
Dolichopodidae	*Chrysotimus molliculus*				0.64	**	Declining
Dolichopodidae	*Chrysotimus flaviventris*				0.78	***	Declining
Empididae	*Rhamphomyia nitidicollis*				0.52	*	Declining
Empididae	*Hilara* sp.				0.76	**	Declining
Hybotidae	*Platypalpus agilis*				0.56	*	Declining
Hybotidae	*Drapetis pusilla*				0.76	***	Declining
Hybotidae	*Platypalpus clarandus*				0.74	***	Declining
Muscidae	*Helina depuncta*				0.46	*	Declining
Muscidae	*Helina impuncta*				0.74	***	Declining
Periscelidae	*Periscelis piricercus*				0.62	**	Declining
Polleniidae	*Pollenia rudis*				0.63	**	Declining
Rhagionidae	*Rhagio lineola*				0.55	*	Declining
Stratiomyidae	*Eupachygaster tarsalis*				0.52	*	Declining
Tachinidae	*Prosethilla kramerella*				0.52	*	Healthy

Note: Flor.: Floricolous species in yellow, Pred.: Predator or parasitoid in red, Sx.: Saproxylic species in brown. *<0.05; **<0.01; ***<0.001.

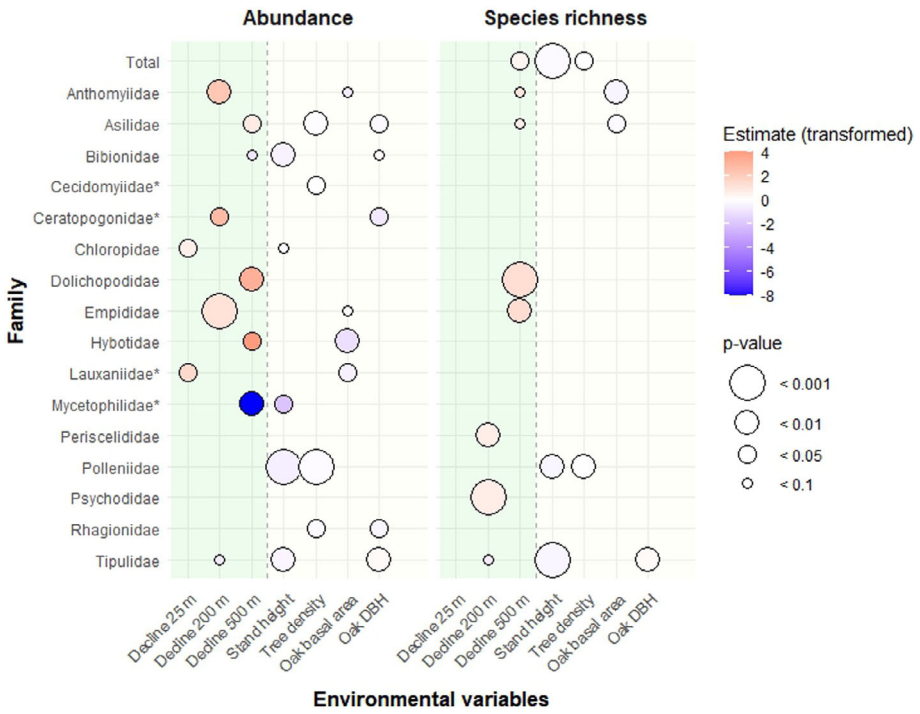


FIGURE 5 Effect of decline severity estimated at three spatial scales (25, 200 and 500 m) on the abundance and species richness of Diptera families collected in green multi-funnel traps, set in oak canopies. Dot colours indicate the value of model estimates; dot sizes increase as the *p*-values decrease. Families with an * were not or partially identified at genus or species levels during the study and were consequently discarded for the analyses on species richness. Estimates were transformed using a signed square-root transformation to reduce the influence of extreme values while preserving the direction of the effects.

diversity (Anthomyiidae and Asilidae). Consequently, overall species richness in the identified community increased slightly but significantly with increasing decline severity. In contrast, Mycetophilidae were significantly less abundant with increasing decline severity, while Bibionidae and Tipulidae showed a similar though non-significant trend (Figure 5).

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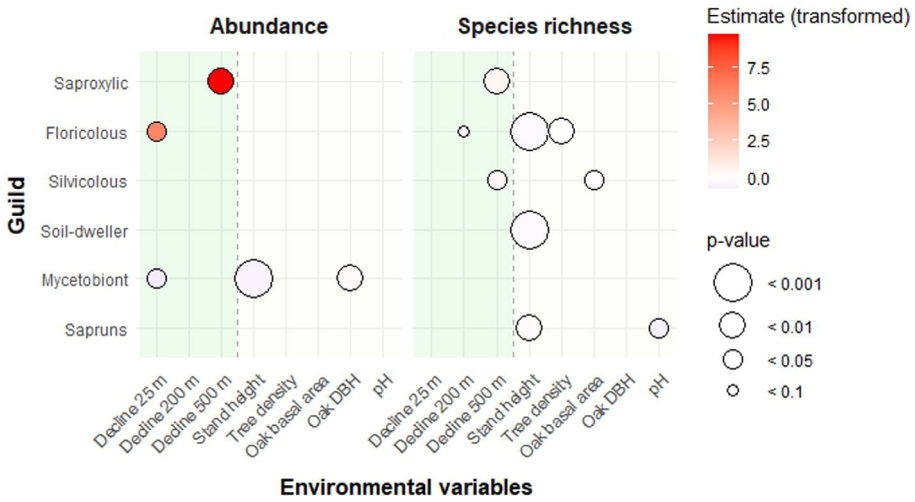


FIGURE 6 Effect of decline severity estimated at three spatial scales (25, 200 and 500 m) on the abundance and species richness of ecological guilds of Diptera collected in green multi-funnel traps, set in oak canopies. Dot colours indicate the value of model estimates; dot sizes increase as the *p*-values decrease. Estimates were transformed using a signed square-root transformation to reduce the influence of extreme values while preserving the direction of the effects.

Stand-related variables such as oak DBH, oak basal area, stand height and tree density had weak negative effects on both the abundance and richness of 11 families (Figure 5). The only exception was a positive effect of oak DBH on both the abundance and richness of Tipulidae. At the community scale, stand height and tree density negatively affected species richness in the identified assemblage.

Effects of decline severity and other environmental variables on the ecological guilds

We estimated that the identified community comprised 298 silvicolous taxa (11,763 individuals), 264 floricolous taxa (6549 individuals), 191 saproxylic taxa (7138 individuals), 134 soil-dwelling taxa (3077 individuals), 17 taxa of sap-run specialists (271 individuals), 13 cavicolous taxa (47 individuals) and 8 mycetobiont taxa (91 individuals) (Table S2). Silvicolous, floricolous and saproxylic taxa were the most species-rich and abundant guilds, whereas cavicolous, mycetobiont taxa and sap-run specialists were comparatively rare in both richness and abundance. Of the seven guilds considered, four were significantly influenced by decline severity (Figure 6). Saproxylic species became more abundant and diverse as decline severity increased. Floricolous species were also more abundant and showed a tendency towards greater diversity, while silvicolous species were only more diverse. In contrast, mycetobiont species were less diverse under higher local decline severity.

Other environmental variables also shaped guild composition (Figure 6). Stand height negatively affected the abundance of mycetobiont species and the richness of floricolous and soil-dwelling species, but positively influenced sap-run specialists. Higher tree density and greater oak basal area reduced the diversity of floricolous and silvicolous species, respectively. Oak DBH promoted the abundance of mycetobiont species, whereas soil pH negatively affected the diversity of sap-run specialists.

Effect of decline severity on species co-occurrence

For the co-occurrence analysis, 11,302 species pair combinations among 475 species were included for the overall community. Among these pair combinations, 479 (4.2%) of them were positive and 191 (1.7%) were negative. The remaining 10,632 combinations (94.1%) were random. There were more positive than negative combinations. Similar results were obtained for each category of decline (Figure S1, Table S3), with random pair combinations accounting for approximately 95% of the total number analysed and more positive than negative associations among healthy, moderately declining and declining plots.

DISCUSSION

With 500 taxa identified from only 42% of the specimens, the canopy-dwelling Diptera community proved to be remarkably diverse, reaching levels comparable to those of the two other major insect orders, Hymenoptera and Coleoptera, also sampled within the same project (Le Souchu, Cours, et al., 2024). This finding supports earlier observations suggesting that Diptera may be as, if not more, abundant and diverse as Coleoptera in certain forest microhabitats and ecosystems (Persson et al., 2013; Vanderwel et al., 2006). Fifteen species were newly recorded for France, including one species new to science (*Priscoearomyia bausenbergensis* Reimann & Rulik; Reimann & Rulik, 2024) and four that likely represent undescribed taxa, highlighting the extent to which this hyper-diverse group remains understudied. We did not investigate the diversity of the most abundant and

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species-rich families which harbour a high proportion of dark taxa, such as Chironomidae, Ceratopogonidae, Phoridae and Sciaridae, and only partially explored that of Mycetophilidae (Morinière et al., 2019). Consequently, although we documented a diverse community, the overall diversity of canopy Diptera was likely underestimated in our study. Since the study was originally designed for beetles, the use of green multi-funnel traps was not ideal for Diptera, which are more effectively collected with Malaise, light, pan, or baited traps (Brown, 2020; Scherber et al., 2014). Nevertheless, the taxonomic composition obtained here closely mirrors that reported in other surveys of forest Diptera, where Sciaridae, Phoridae, Empididae, Chironomidae, Mycetophilidae and Hybotidae were also among the most abundant families (Maguire et al., 2014; Mlynarek et al., 2018; Scherber et al., 2014). Families typically associated with the canopy, such as the Anthomyiidae, Dolichopodidae, Lonchaeidae, Periscelididae and Ulidiidae, were also well represented in our sampling (Maguire et al., 2014; Ruchin et al., 2024).

Decline severity emerged as the most influential environmental factor shaping the community structure of canopy-dwelling Diptera, corroborating previous findings observed in other canopy-dwelling taxa (Le Souchu, Cours, et al., 2024; Sallé et al., 2020). Although most Diptera families were unaffected by decline severity, those that did respond generally benefited from higher levels of decline. Consequently, overall community diversity increased slightly but significantly with greater decline severity. The co-occurrence analysis showed that most species pairs were randomly associated at the plot scale, consistent with previous studies (Elo et al., 2021; Gotelli & Ulrich, 2010; Ulyshen et al., 2020). Among the non-random associations detected, positive co-occurrences were more frequent than negative ones, regardless of decline severity. Although forest decline significantly influenced community composition and species richness, it did not appear to affect the proportion of non-random associations. The predominance of positive associations, while unexpected, may be related to processes such as dispersal or environmental filtering (Elo et al., 2021; Lyons et al., 2016). These positive associations would then likely reflect shared environmental preferences rather than direct biotic interactions. However, as these mechanisms were not directly assessed, the underlying ecological processes remain uncertain and should be interpreted with caution.

As expected, greater decline severity was associated with higher abundances of both saproxylic and floricolous species. Species richness also increased, although for floricolous taxa this effect was only evident as a trend. These patterns were likely driven primarily by the increase in abundance and diversity of Anthomyiidae, Empididae, Hybotidae and Dolichopodidae, families whose larvae are often saproxylic predators while most of the adults are floricolous and/or predators (Cumming et al., 2018; Larson et al., 2001; Sinclair & Daugeron, 2017; Ulyshen, 2018). Several species from these four families such as *Chrysotimus molliculus* (Fallén), *Chrysotimus flaviventris* (von Roser), *Hilara* sp., *Drapetis pusilla* Loew and *Platypalpus clarandus* (Collin) were also strongly associated with oak decline. Pulses in abundance of floricolous insect species are frequently observed following forest disturbances (Cours et al., 2023; Fettig et al., 2022; Viljur et al., 2022). Likewise, saproxylic beetles typically respond positively to forest disturbances (Cours et al., 2022, 2023; Thorn et al., 2017; Viljur et al., 2022). Within the frame of this project, pollinivorous and/or nectarivorous Hymenoptera, as well as wood-nesting (i.e., considered as saproxylic) Hymenoptera similarly benefited from oak decline (Le Souchu, Cours, et al., 2024).

Because positive effects of forest disturbances on cavicolous beetles and cavity-nesting Hymenoptera have been reported elsewhere (Cours et al., 2022; Simanonok & Burkle, 2019), we hypothesized that forest decline would also favour guilds associated with tree-related microhabitats linked to decaying trees, such as rot holes and sap runs. The increase in species richness of Psychodidae and Periscelididae with greater decline severity is consistent with this expectation. The sampled Psychodidae community included several species developing in wet cavities (Jaume-Schinkel et al., 2022), whereas all recorded Periscelididae species are known to develop in sap runs (Papp, 1998; Table S2). Notably, one species, *Periscelis piricercus* Carles-Tolra & Paez, was specifically associated with declining stands. However, we did not detect an overall effect of decline on cavicolous or sap-run-associated guilds, suggesting that responses may be taxon-specific and should not be readily generalized at the guild level.

Disturbance-driven shifts in arthropod community composition are largely mediated by changes in resource availability and microhabitat distribution (Cours et al., 2023; Sallé et al., 2021). Forest decline generally increases both the amount and diversity of deadwood resources (Bouget et al., 2024; Cours et al., 2023), while canopy opening following crown dieback may promote the development of the herbaceous layer and increase floral resource availability, as commonly observed after forest disturbances (Burkle et al., 2019; Romey et al., 2007; Simanonok & Burkle, 2019). A preliminary study conducted in 2020 on a subset of 18 stands from the same project showed that deadwood volume and diversity, as well as the diversity and functional richness of tree-related microhabitats, were higher in declining oak stands (Bouget et al., 2024). However, no effect of oak decline was detected on the abundance of tree cavities or sap runs (Bouget et al., 2024). This suggests that certain Diptera families, such as Periscelididae and Psychodidae, may respond to subtle or otherwise undetected changes in tree-associated microhabitats. For instance, Periscelididae are associated with sap exudates and preferentially occur in the canopy (Maguire et al., 2014; Ruchin et al., 2024), and may develop in perched sap exudates—microhabitats that are difficult to detect from ground level.

Alternatively, warmer and more illuminated conditions in more open, declining stands may have promoted Diptera activity, as temperature and light are known drivers of behaviour and abundance in some Diptera taxa (e.g., Martin et al., 1994). In boreal forests, Kazerani et al. (2025) showed that Diptera biomass and diversity can peak in early-successional or open habitats such as clearcuts, suggesting that Diptera may respond more strongly to structural features such as canopy openness and successional stage than to deadwood availability alone. Taken together, these mechanisms indicate that both resource availability and microclimatic changes may have contributed to

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Diptera responses to oak decline, but our results do not allow us to disentangle their relative importance.

The Mycetophilidae departed from the general positive effect of oak decline observed in the other Diptera families. Their larvae are associated either with the fruiting bodies of macrofungi, such as Agaricales, Russulales and Polyporales, or with fungal mycelia in soil litter or deadwood (Jakovlev, 2012). Consequently, some species are linked to wood-degrading fungi and can be considered saproxylic, while others are associated with mycorrhizal fungi and can be classified as mycetobionts under the guild system used in our study. Although Mycetophilidae were abundant in our samples, we identified only a fraction of the collected specimens and thus obtained only a partial picture of their diversity and ecology. It was therefore not possible to determine whether the observed decline was driven by negative effects on mycetobiont and/or saproxylic taxa. Nevertheless, this decrease in abundance mirrors patterns observed in other mycetobionts, suggesting a broader negative response among mushroom-associated taxa. In a previous investigation of tree-related microhabitats associated with oak decline, perennial fungi were found to be more abundant in declining stands, whereas no effect was detected on ephemeral fungi (Bouget et al., 2024). However, since the productivity of mycorrhizal mushrooms partly depends on tree growth (Egli et al., 2010; Olano et al., 2020), the reduced vigour of declining oak stands may have lowered mycorrhizal mushroom biomass. Mycetophilidae are also known to prefer dark, closed forests, moist microhabitats, and are sensitive to drought (Økland et al., 2005). Reduced canopy closure can also affect the ecological succession of wood-colonizing fungi, leading to less diverse fungal communities in advanced decomposition stages (Schreiber et al., 2024). Thus, the more open conditions created by oak decline, along with the progressive loss of thermal buffering and vertical microclimatic stratification (Sallé et al., 2021), and potentially altered fungal resources, may have further contributed to the decline of Mycetophilidae and other mycetobiont Diptera.

While overall Tachinidae abundance was not affected by variation in decline severity, one tachinid species, *P. kramerella*, was significantly associated with healthy stands. The larval hosts of this species are unknown, but since most Exoristinae are parasitoids of caterpillars (Stireman III et al., 2006), it is probable that this species also targets caterpillars. Interestingly, a similar negative effect of oak decline on parasitoid wasps of defoliating or leaf-mining moths has been reported (Le Souchu, Cours, et al., 2024), suggesting potential cascading effects of changes in foliage quality and/or quantity on defoliators and their associated parasitoids.

We expected that increasing decline severity, and the associated progressive opening of the canopy, would affect silvicolous species. However, their species richness increased with decline severity. Because silvicolous species were broadly defined as forest-associated rather than strictly shade-dependent, this pattern does not necessarily indicate sensitivity to canopy opening. Instead, it may reflect changes in habitat conditions associated with decline, such as increased structural heterogeneity or the greater availability of resources linked to declining trees. However, as microclimatic conditions and deadwood availability were not directly measured in this study, these mechanisms remain speculative and should be interpreted with caution.

We investigated the effects of oak decline using satellite-derived data across three spatial scales. The estimated decline severity values spanned a wide range and were strongly correlated across all scales. Variables measured at larger spatial scales (i.e., 200 and 500 m) proved to be the most explanatory in our analyses. Similarly, Cours et al. (2022) highlighted the importance of landscape-scale processes when examining the effects of forest dieback on local saproxylic beetle assemblages. They identified a 200 m spatial buffer as the most relevant scale for species richness in most saproxylic beetle guilds, consistent with previous estimates of 40–120 m for the same group (Seibold et al., 2017). The ability of insects to respond to large-scale environmental variation generally depends on their dispersal capacity. For instance, dispersal ranges in saproxylic insects may vary from a few meters to several hundred kilometres depending on the spatial and temporal variability of trophic resources and microhabitats, feeding strategies and biotic interactions (Feldhaar & Schauer, 2018). Many saproxylic beetle species are therefore considered highly mobile and only weakly dispersal-limited (Cours et al., 2022). However, the dispersal behaviour and capacities of most Diptera taxa collected in this study remain poorly known and are likely highly variable across taxa and ecological guilds. This currently limits our ability to interpret scale-dependent effects.

Beyond decline severity, other stand-related variables also influenced the abundance and diversity of Diptera families and ecological guilds. With few exceptions, we detected predominantly negative effects of increasing stand height, tree density, oak basal area and DBH. These negative effects may reflect shifts in resource distribution and microclimatic conditions. For instance, taller and denser stands with closed canopies may be less favourable to the herbaceous layer, thereby reducing the availability of floral resources for floricolous species. Conversely, warmer spring and summer conditions in more open stands may enhance larval development and increase the abundance of soil-dwelling species, as previously observed for ground-nesting wasps and beetles (Le Souchu, Cours, et al., 2024; Sallé et al., 2020). Such conditions may also stimulate overall flying activity and the likelihood of insect capture. Alternatively, the negative effects could also be explained by changes in trapping efficiency. For example, the negative effect of stand height on soil-dwelling (e.g., Bibionidae, Tipulidae), floricolous (e.g., Polleniidae) and mycetobiont families may simply reflect the fact that traps were suspended higher in taller stands, thereby reducing their efficiency in capturing species reliant on ground-level resources. This is consistent with observations that Tipulidae are more abundant in lower forest strata (Ruchin et al., 2024). Similarly, the negative effects of tree density, oak basal area and DBH may arise from lower trap detectability in denser stands or in stands with larger trees. Further investigations into the environmental requirements of these species, the vertical stratification of the respective guilds and families, and the attractiveness of green multi-funnel traps would be needed to properly test these hypotheses.

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CONCLUSION

The community of canopy-dwelling Diptera responded significantly to oak decline, highlighting their sensitivity to changes in forest condition. Responses observed at the taxon or family level may be explained by two non-exclusive mechanisms: shifts in microclimatic conditions, with sunnier and warmer environments promoting Diptera activity, and changes in the availability of decline-related resources and habitats such as deadwood, cavities, sap runs and floral resources, which support a variety of Diptera guilds. However, the relative importance of these two mechanisms could not be disentangled in this study. These results nevertheless underscore the potential of forest Diptera as bioindicators of forest ecosystem change.

Despite this potential, two major challenges must be addressed before Diptera can be used effectively for bioindication in forest ecosystems. First, a persistent taxonomic challenge remains: the identification of several groups is complex, and community-level studies of forest Diptera are still scarce. Second, ecological knowledge is limited for many species, which currently restricts the development and interpretation of guild-based analyses.

In addition, the absence of a comprehensive baseline of the local insect fauna limits our ability to interpret community changes along the stand-decline gradient. Establishing such a reference would help clarify what is being lost and what is being gained as forests shift from healthy to declining conditions. Finally, several key insect guilds that play crucial roles in forest dynamics, such as parasitoids, predators, phytophagous and saprophagous taxa, remain insufficiently sampled (but see Sallé et al., 2020; Le Souchu, Cours, et al., 2024). Expanding future analyses to include these groups would provide a more complete understanding of the effects of forest decline on native insect communities.

AUTHOR CONTRIBUTIONS

Anastasia Paupe: Investigation; validation; data curation; writing – review and editing. **Elodie Le Souchu:** Investigation; writing – review and editing; formal analysis; visualization; methodology. **Christophe Bouget:** Conceptualization; methodology; supervision. **Christophe Daugeron:** Writing – review and editing; investigation; validation. **Clovis Quindroit:** Validation; writing – review and editing; investigation. **Pierre-Yves Gloaguen:** Validation; writing – review and editing; investigation. **Patrick Grootaert:** Investigation; validation; writing – review and editing. **Mike Ashworth:** Investigation; validation; writing – review and editing. **Paul L. T. Beuk:** Investigation; validation; writing – review and editing. **Christophe Lauriaut:** Investigation; validation; writing – review and editing. **Valery O. Korneyev:** Investigation; validation; writing – review and editing. **Elena P. Kameneva:** Investigation; validation; writing – review and editing. **André Reimann:** Investigation; validation; writing – review and editing. **Julia J. Mlynarek:** Investigation; validation; writing – review and editing. **Jocelyn Claude:** Investigation; validation; writing – review and editing. **Pierre Tillier:** Investigation; validation; writing – review and editing. **Vivian Feng:** Investigation; validation; writing – review and editing; methodology. **Dalton de Souza Amorim:** Investigation; validation; writing – review and editing. **Rudolf Meier:** Investigation; validation; writing – review and editing; resources. **Ximo Mengual:** Investigation; validation; writing – review and editing. **Tatiana Kúdelová:** Investigation; validation; writing – review and editing. **Thomas Lebard:** Investigation; validation; writing – review and editing. **Anders Lindström:** Investigation; methodology; validation; writing – review and editing; resources. **Tobias Lilja:** Investigation; methodology; validation; writing – review and editing. **Santiago Jaume-Schinkel:** Investigation; methodology; validation; writing – review and editing. **Jonas Mortelmans:** Investigation; validation; writing – review and editing. **Marija Ivković:** Investigation; validation; writing – review and editing. **David Brice:** Investigation; validation; writing – review and editing. **Bruno Righetti:** Investigation; validation; writing – review and editing. **Jean-Paul Haenni:** Investigation; validation; writing – review and editing. **Pauline Bierre:** Investigation; visualization; writing – review and editing; formal analysis. **Cécile Vincent-Barbaroux:** Investigation; methodology; writing – review and editing. **Aurélien Sallé:** Conceptualization; investigation; funding acquisition; writing – original draft; methodology; visualization; formal analysis; project administration; supervision; resources; data curation.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available at ZENODO https://zenodo.org/records/18929240 (Sallé, 2026).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Stands dendrometric characteristics and decline severity values (DSVs) of the sampled plots. Long: longitude, Lat.: latitude, DBH: diameter at breast height.

Table S2. List of taxa collected in 2021 in the canopy of six oak forests in France. The abundance of each taxon, the identifier and the ecological guild to which it belongs are indicated.

Table S3. Summary of results from co-occurrence analysis for the overall Diptera community and each decline category.

Figure S1. Co-occurrences of Diptera species for the overall community (top left) and within healthy (top right), moderately declining (bottom right) and declining (bottom right) contexts, showing negative, positive and random situations.

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